Institute of Technical Education and Research
ITER, SOA

Software Requirements Specification (SRS)

For

Signix

AI-Powered Accessibility Platform for Hearing and Speech Impaired Communication

By

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1. Introduction & Purpose

Signix is an AI-powered accessibility platform designed to improve communication for people with hearing and speech impairments. The system provides real-time translation between sign language, speech, and text while also offering sign language learning resources and emergency communication support. This Software Requirements Specification (SRS) defines the functional and non-functional requirements of the proposed system.

2.Scope

In Scope:

- User Registration and Login
- User Profile Management
- Sign Language Learning Module
- Text-to-Sign Translation
- Speech-to-Text Conversion
- Text-to-Speech Conversion
- AI-based Sign Language Recognition
- Communication Dashboard
- User History
- Admin Dashboard

Out of Scope:

- Medical diagnosis
- Regional sign language support (Phase 2)
- Offline AI processing
- Hardware devices such as smart gloves

3.Functional Requirements

Module 1: Authentication

- Register, login, logout and password recovery.

Module 2: User Profile

- Manage profile, view history, update settings.

Module 3: AI Translation

- Sign→Text
- Text→Sign
- Speech→Text
- Text→Speech
- Real-time translated output.

Module 4: Admin

- Manage users, content, reports and feedback.

4. Non-Functional Requirements

- **Performance:** Average response time shall not exceed 3 seconds.
- **Security:** Encrypted passwords and role-based authentication.
- **Usability:** Responsive and accessible UI for web and mobile.
- **Reliability:** System availability of at least 99% during deployment.

5. SystemArchitecture&TechnologyStack

Architecture: Client → Frontend → Backend API → AI Engine → Database
Frontend: React.js (Web),[Flutter(Mobile){Future Reference}]
Backend: Django
Database: MySql
AI/ML: MediaPipe Hands, TensorFlow, OpenCV
Others: Google Text-to-Speech API, Speech Recognition API, REST APIs

6.Data Requirements

Entity	Key Attributes
User	User ID, Name, Email, Password, Role
Sign Translation	Translation ID, Input, Output, Date
Admin	Admin ID, Username, Password

7.Use Cases

Use Case	Actor	Description
Register	Guest User	Creates a new user account in the system.
Login	User, Admin	Authenticates the user to access the system.
Learn Signs	User	Provides learning resources for sign language.
Translate Sign to Text	User	Converts sign language gestures into text.
Translate Text to Sign	User	Converts text into sign language gestures.
Speech to Text	User	Converts spoken words into text.

Text to Speech	User	Converts text into spoken audio.
Manage Users	Admin	Manages user accounts and system records.

8. Assumptions & Constraints

Assumptions:

- Users have access to a device with a working webcam.
- A stable internet connection is available while using the application.
- Camera and microphone permissions are enabled.
- The AI model has been trained with the required sign language dataset.
- The user performs gestures in a well-lit environment for better recognition.

Constraints:

- The accuracy of sign recognition may decrease in poor lighting or with a low-quality camera.
- The system can recognize only the signs that are included in the trained dataset.
- An internet connection is required to access the web application.
- The project is developed within the time and resource limits of the academic schedule.

9. Expected Outcomes

Signix is expected to make communication easier between hearing-impaired individuals and the general public by providing a simple and effective way to translate sign language into text and speech. The platform will also help users learn sign language, improve accessibility, and provide support during emergencies through the SOS feature. Overall, the system aims to reduce communication barriers and create a more inclusive digital environment where everyone can interact more easily.